



Graduation Regulations for the **Bachelor's project** Artificial Intelligence – University of Groningen

1 INTRODUCTION

The Bachelor's project is the final phase of the Bachelor's degree programme. It has three components: teaching in the form of seminars, the project itself and the related Bachelor's thesis.

These Graduation Regulations deal briefly with all the important points relating to the Bachelor's project. Further information is available from the Bachelor's project coordinator during the Bachelor's project seminars.

1.1 Starting points

Students of Artificial Intelligence in the final phase of their Bachelor's degree programme should have two types of interests and skills: Empiricism and Implementation, embedded in Theory. The main focus of these interests varies from person to person. In carrying out the Bachelor's project, the emphasis is on one of these aspects, while an attempt must also be made to include the other aspect.

The starting point for the Bachelor's project is that it must be academic in nature. This means that there must always be a research question, which must be as concrete as possible, so that it is possible to obtain results within the semester and write a report (the Bachelor's thesis) on them. The report should be written in English.

The subject of the project depends on the availability of Bachelor's projects, current research themes and the student's area of interest.

1.2 Position of the Bachelor's project

AI Bachelor's projects are part of the AI department's own research programme. The projects are supervised by a staff member (lecturer) of the department. Depending on the topic of the Bachelor's project, this member of staff may also delegate the supervision to another academic staff member (e.g. a PhD student). However, the member of staff remains responsible for the progress and the academic standard of the Bachelor's project. There may also be a second supervisor.

Internal Bachelor's projects may fall within the following research themes: Autonomous Perceptive Systems (incl. Robotics and Handwriting Recognition), Multi-agent Systems and Knowledge Technology, and Cognitive Modelling (incl. Cognitive Engineering and Language). In some cases this research is carried out within research institutes which, through staff members, are indirectly affiliated to AI (e.g. psychophysical and biophysical research (hearing) and language research).

1.3 Conditions

The conditions for beginning a Bachelor's project are set out in the Teaching and Examination Regulations (OER) for the Bachelor's degree programme offered by Artificial Intelligence. Students are required to have obtained their propaedeutic certificate and to have passed the Statistics and Research Methods course units.

2 STRUCTURE OF THE BACHELOR'S PROJECT

2.1 Work procedure

For one semester, the student attends compulsory seminars, does the related assignments, carries out the Bachelor's project itself and writes a Bachelor's thesis on it. The Bachelor's project is supervised by a researcher. The student and supervisor meet on a regular basis to discuss the conducting of the research and the statistical and other processing of the results.

2.2 Preparation for the Bachelor's project

In principle, students choose a Bachelor's project from the projects offered. Steps are taken to ensure that a wide range of Bachelor's projects is offered. In the case of a Bachelor's project there is usually already a research question or problem definition, but the student may formulate his/her own research question in consultation with the supervisor.

2.3 Tutorials and assignments

For one semester, students attend tutorials (seminars) and carry out the related assignments. The lectures deal with theory-related subjects such as research methodology (including statistics) and philosophy of science, as well as academic skills such as literature research (library skills is part of the Bachelor's project) and writing academic/scientific reports. In the assignments, students apply what they have learned to their own research. Together, the assignments are the first step towards writing the introduction to the report.

Students are expected to make an active contribution during seminars; attendance is therefore compulsory.

3 THE BACHELOR'S PROJECT

The Bachelor's project is worth 10 ECTS credit points and must be completed within one semester. After this period the student is no longer entitled to supervision.

3.1 Supervision

The Bachelor's project is supervised by a researcher.

The student is entitled to 10 hours of supervision during the semester in which he/she is working on the project. The student and supervisor may agree between themselves on how to schedule the supervision sessions: a half-hour supervision session every week or, for example, a one-hour session every two weeks. The guideline is that students discuss their project with their supervisor at least once every two weeks.

In principle, there will be no further supervision after this semester (no meetings, no feedback on submitted work). The report on the Bachelor's project submitted after the semester will be assessed only as the final version. The final version of the report will not necessarily be approved (it can still be rejected at this stage)!

3.2 Contract

Before the project begins, the student and supervisor make agreements about the nature and frequency of the supervision, about the work to be submitted, the schedule and deadlines. These agreements are set down in a contract that is signed by the student and the supervisor.

3.3 Evaluation

After two months, the progress of the project is evaluated. Both parties have the right to end the supervisory relationship or make further agreements about the nature and/or other aspects of the supervision. The date of the evaluation is specified in the contract.

3.4 Confidentiality obligation

Results (data, software) may only be distributed externally with the permission of the supervisor.

A confidentiality requirement may apply to projects carried out at companies. Students who are working on such projects should be aware that this type of confidentiality requirement may make it impossible to assess the academic quality of the project. Students who wish to carry out a project of this type must therefore make appropriate agreements with their supervisor.

4 COMPLETION OF THE BACHELOR'S PROJECT

4.1 Bachelor's project report / thesis

The project is rounded off with a report. In principle, students write the report individually. Other arrangements are only possible if this is not appropriate for the nature of the project. In such cases, it must be possible to identify the individual contributions, and all the students involved in the project must contribute equally to the report. It is usually *not* possible to write a joint report, but only if this has been jointly agreed by the students and supervisor.

There is a template for the Bachelor's thesis. Students must submit their thesis in accordance with the instructions in the Report Template. The template can be found on Nestor.

The first version of the Bachelor's thesis usually requires further improvement, and changes need to be made. The Bachelor's project report is not approved until the supervisor(s) consider(s) it to be of the required standard (see also under 'Assessment').

4.2 The Bachelor's symposium

Towards the end of the project, the student gives a presentation on the project during the Bachelor's symposium. The student must register for the symposium in advance, and submit a summary (abstract) of no more than 200 words about two weeks in advance. This summary will be used in the invitation to other researchers and students.

The supervisor should attend the presentation because it counts towards the final assessment. The *student* is responsible for inviting the supervisor, and should do so in good time in order to be sure that the supervisor is able to attend.

The length of the presentation is 10 minutes. If more than one student is working on a particular project, they may also give a joint presentation, or individual presentations that are coordinated with each other (several projects in succession).

4.3 Assessment

An *Assessment Form* is used for assessing the Bachelor's project. The final assessment of the Bachelor's project is based on four components:

1. the quality of the presentation of the research results (the presentation during the Bachelor's symposium)
2. the quality of the work performed (experimenting, programming) during the project (effort, independence, etc.)
3. the quality of the thesis (structure, quality of the analyses, etc.)
4. the academic quality of the thesis (publishability, technical and methodological quality, etc.).

One point may be deducted from the final mark for the Bachelor's project if the student has not completed the assignments satisfactorily, has not made an active contribution to the seminars and/or has not met the attendance requirement.

4.4 Completion of the Bachelor's project

In order to receive a grade for the Bachelor's project, the student must submit the Bachelor's thesis (in PDF format) to the University repository (library database) and is responsible for providing the supervisor with a hard copy of the report.

4.5 Completion of the Bachelor's degree programme

Completing the Bachelor's project is not the same thing as completing the Bachelor's degree programme. Students must apply for the Bachelor's degree certificate themselves. More information about this can be found on the internet/Nestor, or with the academic advisor.